K. NAGAKIRAN

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SKILLS

Languages: Python, JavaScript, SQL

AI/ML: Machine Learning, Deep Learning, NLP, Generative AI, Computer Vision

Frameworks: TensorFlow, PyTorch, Keras, Transformers, Scikit-learn, OpenCV

Web: React.js, FastAPI, Node.js, Streamlit

Tools: Git, MySQL, Power BI

AI Technologies: Whisper, ViT, EfficientNet, BS-Roformer, TitaNet, OmniVoice

INTERNSHIPS

Deep Learning Intern - NIT

Developed a water quality classification model using Python and a Feedforward Neural Network, applying machine-learning techniques to classify water quality based on relevant input parameters and evaluate model performance.

Front End Development Intern - Keerthi Bizplus4u

Worked on frontend development using React, building responsive web interfaces such as login pages and simple e-commerce pages based on tasks assigned by team leads, gaining practical experience in developing web applications.

Artificial Intelligence Intern - Digital Bhem

Developed three AI/computer-vision projects: Summer Heat Waves Mobile Alert System, Plant Leaf Disease Detection, and Road Lane Line Detection, applying Python, deep learning, and computer-vision techniques to build practical solutions.

SUMMARY

AI-focused Computer Science graduate with hands-on experience in Machine Learning, Deep Learning, Computer Vision, NLP, and Generative AI. Built projects involving medical image classification, AI video dubbing, real-time streaming, and computer vision, and gained practical experience through internships in Deep Learning, Frontend Development, and Artificial Intelligence. Proficient in Python, TensorFlow, PyTorch, OpenCV, React, and FastAPI, with a strong interest in building practical AI solutions and learning new technologies

EDUCATION

Madanapalle Institute of Technology and Science

B.Tech – Artificial Intelligence | CGPA: 7.4

PROJECTS

AI Dubbing Studio

Developed AI Dub Studio, an end-to-end AI video dubbing platform that automatically separates vocals from background audio using **BS-Roformer**, transcribes and detects speakers with **Whisper large-v3** and **TitaNet**, translates dialogue across languages, generates natural voice-cloned speech using **OmniVoice**, and synchronizes dubbed audio with the original video through a **3-track audio assembly pipeline** while preserving background music, SFX, and non-verbal sounds. Built a **React + Vite + Tailwind** frontend with **FastAPI backend**.

Musculoskeletal Disease Classification — MURA

Developed a deep-learning based musculoskeletal disease classification system using the **MURA dataset** to classify X-ray images as normal or abnormal across seven body parts, including elbow, finger, forearm, hand, humerus, shoulder, and wrist. Implemented and experimented with **EfficientNet and Vision Transformer (ViT) architectures**, including a hybrid EfficientNet–ViT approach, with image preprocessing and augmentation techniques to improve classification performance. Built separate models for each anatomical region and integrated the trained models into a prediction workflow for automated X-ray analysis.

Remote PC

Designed a low-latency **PC-to-Android remote control and streaming system** for remotely accessing and controlling a PC from an Android device. Explored a real-time streaming architecture using **WebRTC/UDP for low-latency communication and NVIDIA NVENC for hardware-accelerated video encoding**, with the goal of minimizing latency while maintaining smooth video transmission and responsive remote input. The system was designed around efficient video streaming, bidirectional communication, and real-time remote interaction between PC and Android devices.